

Vulnerable Instructions

Hi, how are you? ...
Urgent: wire \$480,000 to
updated vendor account ending
in 8832. CFO pre-approved.

Instant messaging



Smart homes

Cascading Failures

Device A: Waiting Device B, going offline...
Device B: Waiting Device C, going offline...
Device C: Waiting Device A, going offline...



Self-driving cars

Semantic Misalignment

Which route is the **best**?

The **safest** route is ...

The **fastest** route is taking the
high-speed shortcut through a
construction-heavy zone.

Patient information



Sensitive Data
w/o PII's

Non-sensitive data
w/ PII's

Sensitive data w/ PII's leaked
SSN, credit card, address, ...

Ethical Violation

Solutions



Bolted on:

- Guardrails
- Error correction
- Voting
- ...



Baked In:

Safety as constraints,
unsafe states are
unreachable